11. Sequences and Series

11.3. Integral Test for Series

11.3 Learning Objectives
- Compare the graph of a series to an integral
- Determine whether a series diverges or converges using the Integral Test
- Bound the remainder of a series

11.3.1. Integral Test

In most cases, we cannot easily determine the actual value that a series converges to. However, we can often determine whether a series is convergent or divergent. In this section, we will begin to investigate methods used to determine whether or not a series is convergent.

Example 11.3.1

Consider the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$. Attempt to use the sequence of partial sums to discuss the convergence of this series.

Solution

The terms in the series have the sequence

$$(a_n)_{n=1}^{\infty} = \left(\frac{1}{1}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{4}} = \frac{1}{2}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{6}}, \ldots\right)$$

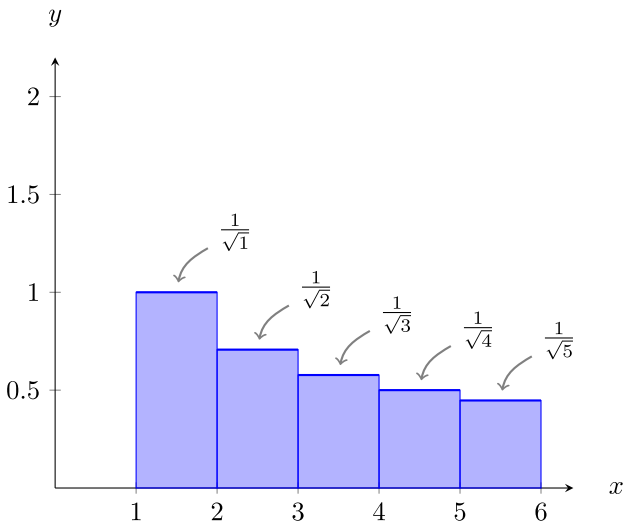
leading us to the sequence of partial sums

$$(s_n)_{n=1}^{\infty} = \left(1, 1 + \frac{1}{\sqrt{2}}, 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}}, \ldots\right)$$

and this is just gross and difficult to work with.

With the harmonic series we could do some clever bounding and algebra, but those are typically difficult techniques to apply in general. Instead let's see if we can approach this from a different angle.

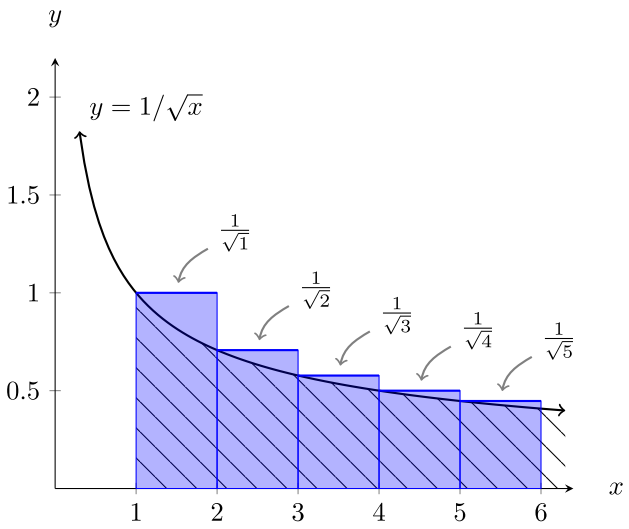
Observe we can "graph" the sequence like a step function. We can construct a bunch of rectangles of width 1 and height $a_n = \frac{1}{\sqrt{n}}$



and the area of all the rectangles is the series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}.$$

A lower bound for the area under this curve is the integral of the function that passes through the left end points of our rectangles: $y = \frac{1}{\sqrt{x}}$! (not factorial, just excitement)



We can use the convergence or divergence of improper integrals that directly correspond to the series.

In this case, we already know that $\int_1^{\infty} \frac{1}{\sqrt{x}}\,\mathrm{d}x$ diverges since it is of the form $\int_1^{\infty} \frac{1}{x^p}\,\mathrm{d}x$ with $p \leq 1$.

However, visually we can see that $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} > \int_1^{\infty} \frac{1}{\sqrt{x}}\,\mathrm{d}x$, since the area of each of the rectangles is like a left Reimann sum for the integral and the function is decreasing.

Thus, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverges.

Example 11.3.2

Determine the convergence of the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

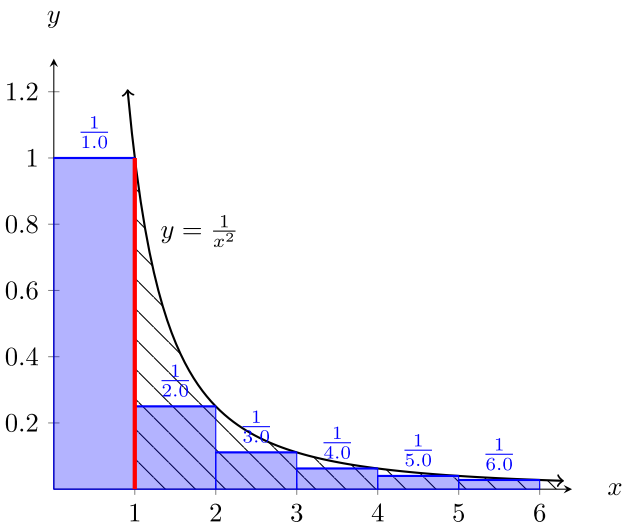
by comparing to a known integral.

Solution

Let's start with a picture, in this case, we suspect that this series converges, so we want to find a curve above some rectangles to help us visualize things. We can write

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2}$$

then, the series can be represented by the rectangles to the right of the red line in this picture:



Notice $\sum_{n=2}^{\infty} \frac{1}{n^2} < \int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x$. Since $\int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x$ is of the form $\int_1^{\infty} \frac{1}{x^p}\,\mathrm{d}x$ and $p > 1$, this integral converges, so the series

$$\sum_{n=2}^{\infty} \frac{1}{n^2}$$

converges. Then, since we only need to add 1 (which is finite) we know

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges as well.

Breaking off the 1 at the start, we can do a similar trick with any convergent series, so we don't have to do this in the future. We can generalize these results with the following test for convergence.

Theorem 11.3.1 The Integral Test

Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, and decreasing function of n for all $n \geq N$ (where N is a positive integer). Then,

(i) If $\int_N^{\infty} f(x)\,\mathrm{d}x$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is a convergent series.

(ii) If $\int_N^{\infty} f(x)\,\mathrm{d}x$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is a divergent series.

Warning

- This does not tell you what value the series converges to!!!!!!
- This test does not always hold without all 3 of the assumptions. f must be continuous, positive, and decreasing to use this test. If you don't check these conditions you can't use the integral test.

We have already determined for which values of p the integral $\int_1^{\infty} \frac{1}{x^p}\,\mathrm{d}x$ converges and diverges. So...

Corollary 11.3.1 p-Series Test

The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$,

(i) converges if $p > 1$

(ii) diverges if $p \leq 1$

Tip

We are starting to build up a few test we can use for series convergence, and it will begin to be a challenge to identify which to use. For $\sum_{n=1}^{\infty} a_n$ so far we can use

- the divergence test (if $(a_n)_{n=1}^{\infty}$ doesnt converge to 0 then the series diverges)
- Think about the partial sums (usful for telescoping series and things that easily simplify when added together)
- Geometric Series test (is it a geomatric series with $r < 1$?)
- p-series test (for $a_n = \frac{1}{n^p}$ type things)
- integral test (for things that are easy/possible to integrate using our integration techniques)

We will add to this list, so keeping a reference of each test (and a short description of when to use them) easily accessible in your notes will be good.

Example 11.3.3

Determine whether the series $\sum_{n=1}^{\infty} \frac{\ln(n)}{n}$ converges or diverges.

Solution

It us often useful to do a quick check that the sequence converges to 0 before progressing:

$$\lim_n \to \infty \frac{\ln(n)}{n} \overset{\text{LH}}{=} \frac{\frac{1}{n}}{1} = 0$$

So the series is not obviously divergent. We still don't know if it converges or diverges.

Next, we run through our other tests. It won't be easy to simplify the partial sums, and this is not a p-series. So we try the integral test. Let

$$f(x) = \frac{\ln(x)}{x}.$$

Notice this is something we can integrate, let's see if it converges or not: $u = \ln(x)$ $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$ so

$$\begin{aligned}
\int_1^\infty \frac{\ln(x)}{x}\,\mathrm{d}x &= \lim_{t\to\infty} \int_1^t \frac{\ln(x)}{x}\,\mathrm{d}x \\
&= \lim_{t\to\infty} \int_{u(1)}^{u(t)} u\,\mathrm{d}u \\
&= \lim_{t\to\infty} \frac{u^2}{2} \Big|_{u(1)}^{u(t)} \\
&= \lim_{t\to\infty} \left(\frac{1}{2}(\ln(x))^2\right) \Big|_1^t \\
&= \lim_{t\to\infty} \left(\frac{1}{2}(\ln(t))^2 - 0\right) \\
&= \infty
\end{aligned}$$

Thus, **by the Integral Test**, the series diverges.

Warning

- You will lose points for not using proper limit notation on improper integrals in this seciton.
- You will also lose points for not stating which series convergence test you are using. These might be good things to make flashcards of and study as we move towards the exam.

11.3.2. Remainder Estimate for Integral Test

Suppose a series converges, and we want to estimate its value. How can we do this?

- If a series converges, then, by definition (even if it is hard for us to calculate directly), its value is the limit of the sequence of partial sums. So, we can take a sufficiently long partial sum to approximate the value. The larger n we use for the nth partial sum, the better the estimate, since we're adding more of the terms.

Theorem 11.3.2 **Remainder Estimate for Integral Test**

Suppose $f(n) = a_n$, where f is a continuous, positive, decreasing function for $x \geq N$ and $\sum_{n=1}^{\infty} a_n$ is convergent.

The remainder, R_n, is the error made when s_n, the sum of the first n terms, is used as an approximation to the total sum, s. We define

$$R_n = s - s_n = a_{n+1} + a_{n+2} + a_{n+3} + \ldots$$

Then, we know that

$$R_n = a_{n+1} + a_{n+2} + a_{n+3} + \ldots \leq \int_n^{\infty} f(x)\,\mathrm{d}x$$

and

$$R_n = a_{n+1} + a_{n+2} + a_{n+3} + \ldots \geq \int_{n+1}^{\infty} f(x)\,\mathrm{d}x$$

So,

$$\int_{n+1}^{\infty} f(x)\,\mathrm{d}x \leq R_n \leq \int_n^{\infty} f(x)\,\mathrm{d}x$$

gives us an error approximation to the estimation s_n.

Example 11.3.4

Approximate the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ by using the 10th partial sum, namely by adding the first 10 terms of the series. How many terms are needed to guarantee an approximation for the series that is accurate to at least 3 decimal places?

Solution

$$\begin{aligned} s_{10} &= \sum_{n=1}^{10} \frac{1}{n^2} \\ &= 1 + \frac{1}{4} + \frac{1}{9} + \ldots + \frac{1}{100} \approx 1.54976773117 \end{aligned}$$

To guarantee our approximation is accurate to at least 3 decimal places, we need the error (or the remainder R_n) to be less than 0.001. We'll see what value of n guarantees $R_n \leq 0.001$ by solving when $R_n < 0.001$. We know by the remainder estimate that

$$R_n \leq \int_n^{\infty} \frac{1}{x^2}\,\mathrm{d}x$$

This integral is

$$\begin{aligned}\int_n^\infty \frac{1}{x^2}\,\mathrm{d}x &= \lim_{t\to\infty}\int_n^t \frac{1}{x^2}\,\mathrm{d}x \\ &= \lim_{t\to\infty} -\frac{1}{x}\,|_n^t \\ &= \lim_{t\to\infty} \frac{1}{n} - \frac{1}{t} \\ &= \frac{1}{n}.\end{aligned}$$

Then, since we know $R_n \leq \frac{1}{n}$, to get $R_n < 0.001 = \frac{1}{1000}$ we need $\frac{1}{n} < \frac{1}{1000}$ or $n > 1000$. So, to **guarantee** that the estimate is accurate up to 3 decimal places we need to use $n > 1000$.

It turns out this is an overestimate of what is actually needed to get an accurate approximation of the series. We only need up to $n = 32$ to get 3 decimals of accuracy in this case. Much less than 1000.

11.3 Section Summary:

- We used integrals to bound series
- We used integrals to bound the error when approximating series with partial sums.